

Technical Appendix: A Practical Sharpe-Ratio Diagnostic for Portfolio Backtests

1 Purpose

This appendix supplies the derivation and supporting tables for the applied manuscript. It is a technical supplement to the practical date-aggregation diagnostic, not a separate asset-pricing claim.

2 Mean-return design-effect intuition

For a selected-row return model $y_{t,i} = \mu + \lambda_t + \epsilon_{t,i}$ with constant selected count $\bar{n}$, same-date intraclass correlation ρ , and independent dates, the row-pooled mean equals the average of date means:

$$\hat{\mu} = \frac{1}{T\bar{n}} \sum_{t=1}^T \sum_{i=1}^{\bar{n}} y_{t,i} = \frac{1}{T} \sum_{t=1}^T \bar{y}_t.$$

The row-naive variance report is $\sigma^2/(T\bar{n})$. The true one-date variance is

$$\text{Var}(\bar{y}_t) = \frac{\sigma^2}{\bar{n}} \{1 + (\bar{n} - 1)\rho\}.$$

Thus the date-level standard error is larger than the row-naive standard error by

$$\sqrt{1 + (\bar{n} - 1)\rho}.$$

In words, additional same-date rows do not create independent time-series observations when they share common date shocks.

3 Sharpe uncertainty inflation factor

Let R_t be the date-level portfolio excess return and $G_t = (R_t, R_t^2)'$. The annualized Sharpe ratio is the smooth functional

$$f(\mu, m_2) = \sqrt{A_f} \frac{\mu}{(m_2 - \mu^2)^{1/2}},$$

where A_f is the source-specific annualization factor. Under standard delta-method regularity conditions, the date-level long-run variance of the Sharpe estimate is based on the long-run covariance matrix of G_t . The row-pooled comparator applies the same smooth-functional calculation to selected rows as if they were independent.

The reported Sharpe uncertainty inflation factor is the ratio

$$\text{Sharpe uncertainty inflation} = \frac{\text{Var}_{LR}(\widehat{\text{SR}}_{date})}{\text{Var}_{IID, row}(\widehat{\text{SR}}_{row})}.$$

It is a reporting diagnostic. Rejection decisions in the manuscript use date-level HAC, dependent-bootstrap, fixed-menu, placebo, and turnover-cost checks.

4 Supporting empirical tables

This supplement reports loaded sources in separated groups. The loader provenance is retained in the generated metadata; it is not a comparable pass/fail table. The tables are supporting diagnostics rather than primary evidence; the main text reports the compressed interpretation.

Table 1. Source frequency and annualization metadata.

Source	type	period	A_f	dates
AQR Betting Against Beta, equity factors	single_series_factor	daily	252.000	25281
AQR HML Devil	single_series_factor	daily	252.000	26590
AQR Quality Minus Junk	single_series_factor	daily	252.000	17681
AQR Value and Momentum Everywhere	single_series_factor	monthly	12.000	651
French momentum deciles, rank-threshold panel	panel	daily	252.000	26131
French WML decile benchmark	panel	daily	252.000	26131
French 25 Size/BM, skipped-momentum panel	panel	daily	252.000	3477

Table 2. Panel candidates with row-level signal-return structure.

Panel candidate	SR	HAC p+	Boot p+	RW p	Perm p	Net SR	Full rule 5%	Applicable checks 5%
French momentum deciles, rank-threshold panel	0.447	7.54e-05	2.00e-04	2.00e-04	N/A	0.447	no	yes
French WML decile benchmark	0.497	5.53e-06	2.00e-04	2.00e-04	1.00e-04	0.497	yes	yes
French 25 Size/BM, skipped-momentum panel	0.058	0.412	0.422	0.480	0.943	-0.050	no	no

Table 3. Single-series factor benchmarks subject to time-series inference only.

Single-series benchmark	SR	HAC p+	Boot p+	RW p	Scope
AQR Betting Against Beta, equity factors	0.741	3.08e-13	2.00e-04	2.00e-04	time-series only
AQR Quality Minus Junk	0.536	2.26e-05	2.00e-04	2.00e-04	time-series only
AQR HML Devil	0.336	0.004	0.002	0.002	time-series only
AQR Value and Momentum Everywhere	0.572	1.62e-04	2.00e-04	2.00e-04	time-series only

Note: Time-series only means not eligible for same-date permutation, row-retention diagnostics, or the full panel decision rule.

Table 4. Canonical French momentum benchmark validation.

Candidate	thr	n	start	end	panel SR	direct WML SR	scale
French WML decile benchmark	0.900	26131	1926-11-04	2026-04-30	0.497	0.497	0.500

Table 5. Comparison with common row and date-level inference choices for panel candidates.

Panel candidate	Row-naive p+	Date-IID p+	Date-HAC p+	Moving-block p+	Uncertainty infl.
French momentum deciles, rank-threshold panel	1.70e-04	2.00e-04	7.54e-05	2.00e-04	1.000
French WML decile benchmark	4.85e-04	2.00e-04	5.53e-06	2.00e-04	1.000
French 25 Size/BM, skipped-momentum panel	0.253	0.417	0.412	0.422	9.002

Note: The row-naive column treats selected rows as independent. Date-IID, date-HAC, and block-bootstrap columns operate on the date-level portfolio return series.

Table 6. Selected-count and dependence diagnostics for panel candidates.

Candidate	status	thr	rows	dates	avg names	rho	SE infl.
French momentum deciles, rank-threshold panel	Available	0.700	104524	26131	4.000	-0.138	0.766
French WML decile benchmark	Available	0.900	52262	26131	2.000	-0.583	0.646
French 25 Size/BM, skipped-momentum panel	Available	0.500	45201	3477	13.000	0.667	3.000

Table 7. Row-naive and date-level p-value diagnostics for panel candidates.

Candidate	thr	row t	HAC z	row p+	HAC p+	RW p
French momentum deciles, rank-threshold panel	0.700	3.584	3.790	1.70e-04	7.54e-05	2.00e-04
French WML decile benchmark	0.900	3.299	4.395	4.85e-04	5.53e-06	2.00e-04
French 25 Size/BM, skipped-momentum panel	0.500	0.664	0.222	0.253	0.412	0.480

Table 8. Unclipped and one-sided floored uncertainty-inflation diagnostics for panel candidates.

Candidate	rho	SE infl.	Infl. unclipped	Infl. displayed	ERIR unclipped	ERIR displayed
French momentum deciles, rank-threshold panel	-0.138	0.766	0.587	1.000	1.704	1.000
French WML decile benchmark	-0.583	0.646	0.417	1.000	2.399	1.000
French 25 Size/BM, skipped-momentum panel	0.667	3.000	9.002	9.002	0.111	0.111

Note: Displayed main-text uncertainty-inflation values are floored at one when the equicorrelation diagnostic is non-inflationary; unclipped values remain descriptive diagnostics and are not rejection rules.

Table 9. Sampling-boundary summary for panel candidates.

Candidate	Row p+	Infl.	Max p+	Net SR	Status
French momentum deciles, rank-threshold panel	1.70e-04	1.000	2.00e-04	0.447	Permutation uninformative; remaining applicable checks pass
French WML decile benchmark	4.85e-04	1.000	2.00e-04	0.497	Passes full evaluation
French 25 Size/BM, skipped-momentum panel	0.253	9.002	0.943	-0.050	Fails statistical and cost checks

Note: Displayed uncertainty inflation is a one-sided summary floored at one. Degenerate same-date permutation diagnostics are reported as uninformative rather than converted into failures.

Table 10. Horizon-effect sensitivity for the dynamic Size/BM momentum panel.

L	thr	dates	avg names	gross SR	net SR 5bps	turnover	Sharpe infl.	serial infl.	HAC p+
1	0.500	3497	12.913	0.073	-0.394	0.733	11.498	0.891	0.386
5	0.500	3493	13.000	-0.312	-0.528	0.339	12.880	0.990	0.878
21	0.500	3477	13.000	0.058	-0.050	0.169	12.121	0.933	0.412

Table 11. Primary-threshold dependence-aware Sharpe inference.

Candidate	thr	Method	SR	SE	CI low	CI high	p+	n
AQR Betting Against Beta, equity factors	0.000	block	0.741	0.106	0.533	0.951	2.00e-04	25281
AQR Betting Against Beta, equity factors	0.000	HAC	0.741	0.103	0.540	0.943	3.08e-13	25281
AQR Betting Against Beta, equity factors	0.000	prewhite	0.741	0.104	0.538	0.945	4.38e-13	25281
AQR HML Devil	0.000	block	0.336	0.114	0.105	0.552	0.002	26590
AQR HML Devil	0.000	HAC	0.336	0.126	0.089	0.583	0.004	26590
AQR HML Devil	0.000	prewhite	0.336	0.111	0.119	0.553	0.001	26590
AQR Quality Minus Junk	0.000	block	0.536	0.133	0.276	0.795	2.00e-04	17681
AQR Quality Minus Junk	0.000	HAC	0.536	0.131	0.279	0.794	2.26e-05	17681
AQR Quality Minus Junk	0.000	prewhite	0.536	0.128	0.285	0.787	1.40e-05	17681
AQR Value and Momentum Everywhere	0.000	block	0.572	0.156	0.268	0.882	2.00e-04	651
AQR Value and Momentum Everywhere	0.000	HAC	0.573	0.159	0.261	0.885	1.62e-04	651
AQR Value and Momentum Everywhere	0.000	prewhite	0.573	0.158	0.263	0.882	1.42e-04	651
French momentum deciles, rank-threshold panel	0.700	block	0.447	0.114	0.222	0.667	2.00e-04	26131
French momentum deciles, rank-threshold panel	0.700	HAC	0.447	0.118	0.216	0.679	7.54e-05	26131
French momentum deciles, rank-threshold panel	0.700	prewhite	0.447	0.114	0.224	0.670	4.18e-05	26131
French WML decile benchmark	0.900	block	0.497	0.111	0.275	0.708	2.00e-04	26131
French WML decile benchmark	0.900	HAC	0.497	0.113	0.275	0.718	5.53e-06	26131
French WML decile benchmark	0.900	prewhite	0.497	0.111	0.279	0.715	3.99e-06	26131
French 25 Size/BM, skipped-momentum panel	0.500	block	0.058	0.261	-0.483	0.540	0.422	3477
French 25 Size/BM, skipped-momentum panel	0.500	HAC	0.058	0.260	-0.452	0.567	0.412	3477
French 25 Size/BM, skipped-momentum panel	0.500	prewhite	0.058	0.255	-0.442	0.557	0.410	3477

Table 12. Same-date signal-permutation diagnostics for panel candidates.

Candidate	thr	obs SR	null mean	null sd	p+	status	perms
French momentum deciles, rank-threshold panel	0.000	0.395	0.410	0.039	0.646	Available	10000
French momentum deciles, rank-threshold panel	0.300	0.419	0.434	0.024	0.733	Available	10000
French momentum deciles, rank-threshold panel	0.500	0.419	0.434	0.024	0.733	Available	10000
French momentum deciles, rank-threshold panel	0.700	0.447	0.447	1.11e-16	N/A	Permutation null degenerate	10000
French momentum deciles, rank-threshold panel	0.900	0.497	0.404	0.042	0.015	Available	10000
French WML decile benchmark	0.900	0.497	-0.001	0.098	1.00e-04	Available	10000
French 25 Size/BM, skipped-momentum panel	0.000	0.040	0.080	0.017	0.991	Available	10000
French 25 Size/BM, skipped-momentum panel	0.300	0.034	0.097	0.023	0.997	Available	10000
French 25 Size/BM, skipped-momentum panel	0.500	0.058	0.104	0.029	0.943	Available	10000
French 25 Size/BM, skipped-momentum panel	0.700	0.087	0.137	0.041	0.892	Available	10000
French 25 Size/BM, skipped-momentum panel	0.900	0.054	0.119	0.069	0.832	Available	10000

Note: A near-zero permutation-null standard deviation is reported as N/A because the placebo is uninformative for that threshold, not calibrated evidence for or against predictability.

Table 13. HAC bandwidth, prewhitening, and fixed-b robustness diagnostics.

Candidate	Diagnostic	SR	SE	p+
AQR Betting Against Beta, equity factors	HAC K=auto	0.741	0.103	3.08e-13
AQR Betting Against Beta, equity factors	HAC K=21	0.741	0.119	2.04e-10
AQR Betting Against Beta, equity factors	HAC K=63	0.741	0.130	6.06e-09
AQR Betting Against Beta, equity factors	prewhite	0.741	0.104	4.38e-13
AQR Betting Against Beta, equity factors	fixed-b 0.1	0.741	0.166	0.002
AQR HML Devil	HAC K=auto	0.336	0.126	0.004
AQR HML Devil	HAC K=21	0.336	0.128	0.004
AQR HML Devil	HAC K=63	0.336	0.137	0.007
AQR HML Devil	prewhite	0.336	0.111	0.001
AQR HML Devil	fixed-b 0.1	0.336	0.110	0.004
AQR Quality Minus Junk	HAC K=auto	0.536	0.131	2.26e-05
AQR Quality Minus Junk	HAC K=21	0.536	0.140	6.33e-05
AQR Quality Minus Junk	HAC K=63	0.536	0.147	1.32e-04
AQR Quality Minus Junk	prewhite	0.536	0.128	1.40e-05
AQR Quality Minus Junk	fixed-b 0.1	0.536	0.151	0.002
AQR Value and Momentum Everywhere	HAC K=auto	0.573	0.159	1.62e-04
AQR Value and Momentum Everywhere	HAC K=21	0.573	0.155	1.14e-04
AQR Value and Momentum Everywhere	HAC K=63	0.573	0.144	3.41e-05
AQR Value and Momentum Everywhere	prewhite	0.573	0.158	1.42e-04
AQR Value and Momentum Everywhere	fixed-b 0.1	0.573	0.143	0.004
French momentum deciles, rank-threshold panel	HAC K=auto	0.447	0.118	7.54e-05
French momentum deciles, rank-threshold panel	HAC K=21	0.447	0.120	9.94e-05
French momentum deciles, rank-threshold panel	HAC K=63	0.447	0.122	1.21e-04
French momentum deciles, rank-threshold panel	prewhite	0.447	0.114	4.18e-05
French momentum deciles, rank-threshold panel	fixed-b 0.1	0.447	0.143	0.004
French WML decile benchmark	HAC K=auto	0.497	0.113	5.53e-06
French WML decile benchmark	HAC K=21	0.497	0.116	9.62e-06
French WML decile benchmark	HAC K=63	0.497	0.117	1.01e-05
French WML decile benchmark	prewhite	0.497	0.111	3.99e-06
French WML decile benchmark	fixed-b 0.1	0.497	0.147	0.002
French 25 Size/BM, skipped-momentum panel	HAC K=auto	0.058	0.260	0.412
French 25 Size/BM, skipped-momentum panel	HAC K=21	0.058	0.249	0.409
French 25 Size/BM, skipped-momentum panel	HAC K=63	0.058	0.231	0.401
French 25 Size/BM, skipped-momentum panel	prewhite	0.058	0.255	0.410
French 25 Size/BM, skipped-momentum panel	fixed-b 0.1	0.058	0.172	0.360

Table 14. Decision-rule sensitivity by alpha threshold.

alpha	candidates	full passes	applicable passes
0.010	8	1	6
0.050	8	1	6

Table 15. Holdout and subperiod Sharpe diagnostics.

Candidate	window	SR	n
AQR Betting Against Beta, equity factors	train_70pct	0.887	17696
AQR Betting Against Beta, equity factors	holdout_30pct	0.531	7585
AQR Betting Against Beta, equity factors	quarter_min	0.280	6320
AQR Betting Against Beta, equity factors	quarter_max	1.270	6320
AQR HML Devil	train_70pct	0.465	18613
AQR HML Devil	holdout_30pct	0.120	7977
AQR HML Devil	quarter_min	0.273	6647
AQR HML Devil	quarter_max	0.445	6647
AQR Quality Minus Junk	train_70pct	0.794	12376
AQR Quality Minus Junk	holdout_30pct	0.305	5305
AQR Quality Minus Junk	quarter_min	0.174	4421
AQR Quality Minus Junk	quarter_max	1.083	4420
AQR Value and Momentum Everywhere	train_70pct	0.626	455
AQR Value and Momentum Everywhere	holdout_30pct	0.431	196
AQR Value and Momentum Everywhere	quarter_min	0.372	163
AQR Value and Momentum Everywhere	quarter_max	0.941	163
French momentum deciles, rank-threshold panel	train_70pct	0.615	18292
French momentum deciles, rank-threshold panel	holdout_30pct	0.229	7839
French momentum deciles, rank-threshold panel	quarter_min	0.171	6532
French momentum deciles, rank-threshold panel	quarter_max	1.280	6533
French WML decile benchmark	train_70pct	0.641	18292
French WML decile benchmark	holdout_30pct	0.310	7839
French WML decile benchmark	quarter_min	0.224	6533
French WML decile benchmark	quarter_max	1.370	6533
French 25 Size/BM, skipped-momentum panel	train_70pct	0.088	2428
French 25 Size/BM, skipped-momentum panel	holdout_30pct	-0.007	1049
French 25 Size/BM, skipped-momentum panel	quarter_min	-0.581	853
French 25 Size/BM, skipped-momentum panel	quarter_max	0.210	875

5 Supporting simulation tables

Table 16. Turnover-scaled cost sensitivity and break-even cost.

Candidate	cost bps	turnover	gross SR	net SR	break-even bps
French momentum deciles, rank-threshold panel	0	1.91e-05	0.447	0.447	85893.000
French momentum deciles, rank-threshold panel	1	1.91e-05	0.447	0.447	85893.000
French momentum deciles, rank-threshold panel	5	1.91e-05	0.447	0.447	85893.000
French momentum deciles, rank-threshold panel	10	1.91e-05	0.447	0.447	85893.000
French momentum deciles, rank-threshold panel	25	1.91e-05	0.447	0.447	85893.000
French WML decile benchmark	0	1.91e-05	0.497	0.497	122232.000
French WML decile benchmark	1	1.91e-05	0.497	0.497	122232.000
French WML decile benchmark	5	1.91e-05	0.497	0.497	122232.000
French WML decile benchmark	10	1.91e-05	0.497	0.497	122232.000
French WML decile benchmark	25	1.91e-05	0.497	0.497	122232.000
French 25 Size/BM, skipped-momentum panel	0	0.169	0.058	0.058	2.683
French 25 Size/BM, skipped-momentum panel	1	0.169	0.058	0.036	2.683
French 25 Size/BM, skipped-momentum panel	5	0.169	0.058	-0.050	2.683
French 25 Size/BM, skipped-momentum panel	10	0.169	0.058	-0.157	2.683
French 25 Size/BM, skipped-momentum panel	25	0.169	0.058	-0.480	2.683

Note: AQR factor series are pre-aggregated returns. Constituent-level turnover is not observed in the public series, so turnover-cost diagnostics are reported only for row-level panel candidates.

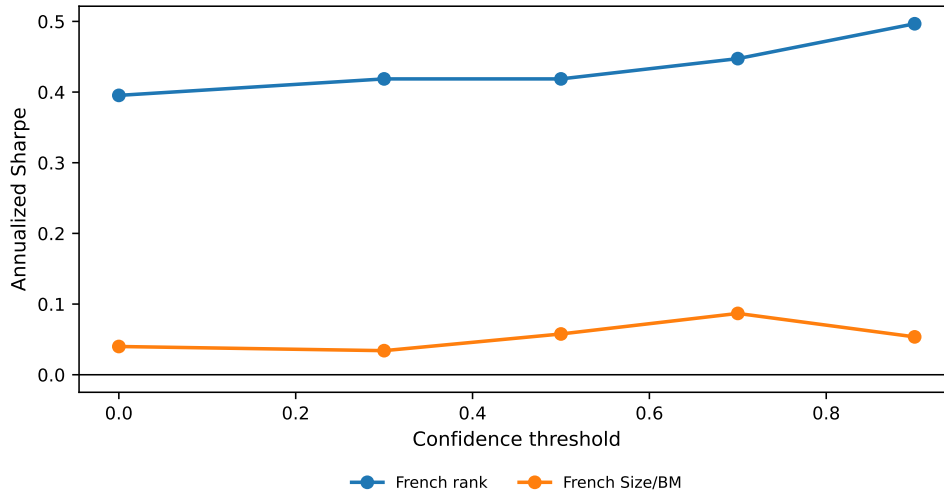


Figure 1. Threshold-profile diagnostics for public candidate panels; each line shows how annualized Sharpe changes when the pre-specified confidence threshold is varied.

Table 17. Monte Carlo design contract.

Design	T	N	Selected count	XS dep.	Serial dep.	Heterosk.	Reps	Resampling	Seed
IID null	1000	50	all rows	$\rho=0$	$AR(1)=0$	none	1000	5000 boot	rep index
Dependent null	1000	50	all rows	$\rho=0.2$	$AR(1)=0.2$	none	1000	5000 boot	rep index
GARCH null	1000	50	all rows	$\rho=0.2$	$AR(1)=0.2$	GARCH(0.05,0.90)	1000	5000 boot	rep index
Positive control	5000	100	signal-selected rows	common loading=0.50	$AR(1)=0.10$	none	100	499 boot	777

Note: All null designs have true Sharpe equal to zero. Positive-control rows use a declared nonzero signal and are included to show that the reporting rule is not mechanically anti-discovery.

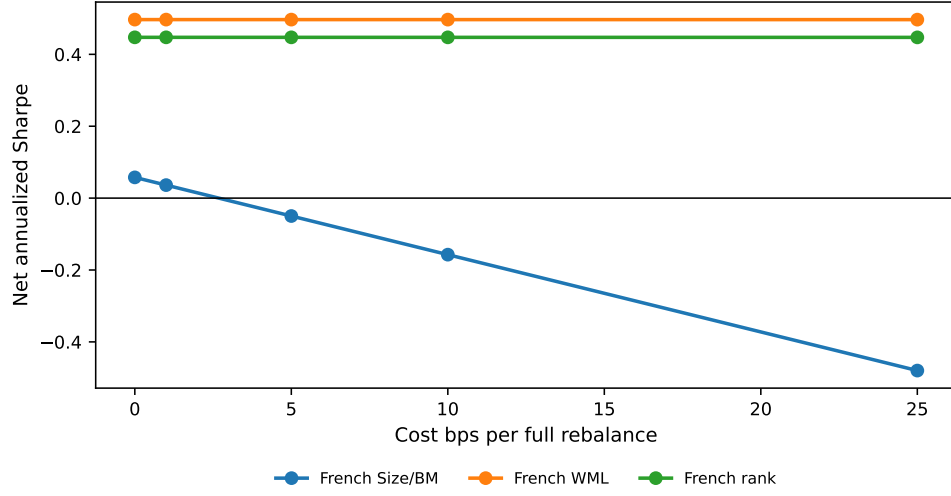


Figure 2. Turnover-scaled cost sensitivity at primary candidate thresholds; positive values indicate that the candidate remains above zero after the displayed linear cost stress.

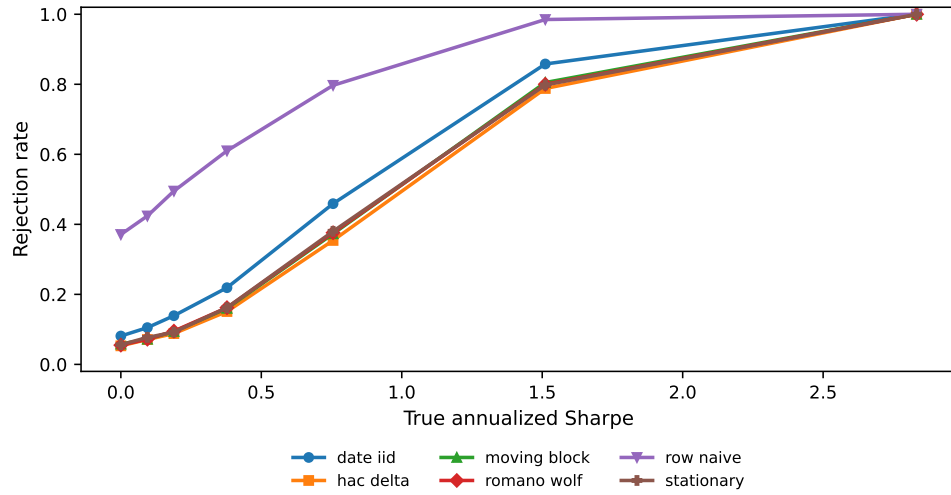


Figure 3. Monte Carlo rejection rates by true annualized Sharpe; the figure separates genuine power from methods that already overreject under null designs with no true edge.

Table 18. Monte Carlo and resampling settings.

Setting	Value
Monte Carlo replications	1000
Bootstrap draws	5000
Permutation draws	10000
Main null panel sizes	T=1000, N=50
Cross-sectional correlation grid	0.000, 0.200, 0.250, 0.300, 0.350, 0.400, 0.500
AR(1) grid	0.000, 0.150, 0.200, 0.300, 0.500, 0.700
GARCH parameter pairs	(0.050, 0.900); (0.100, 0.850); (0.150, 0.800)
HAC bandwidth rule	automatic Bartlett, with fixed-lag sensitivity checks
Moving-block length	ceil(T to the 1/5 times max(D,1)), capped at floor(T/3)
Stationary mean block	same length as moving-block rule
Dependence factor D	(2 rho1/(1-rho1)) to the 2/3 for positive AR(1) rho1; otherwise 1
Seed policy	seed-indexed replications from campaign runner

Table 19. Simulation DGP parameter grid.

DGP	Name	T	N	mu	sigma	AR1	rho	GARCH a	GARCH b	K	regime
01_iid	IID baseline	1000	50	4.00e-04	0.015	0.000	0.000	0.000	0.000	1	no
02_ser_mild	Serial mild	1000	50	4.00e-04	0.015	0.200	0.000	0.000	0.000	1	no
03_ser_strong	Serial strong	1000	50	4.00e-04	0.015	0.500	0.000	0.000	0.000	1	no
04_xs_mild	Cross mild	1000	50	4.00e-04	0.015	0.000	0.200	0.000	0.000	1	no
05_xs_strong	Cross strong	1000	50	4.00e-04	0.015	0.000	0.500	0.000	0.000	1	no
06_both_mod	Both moderate	1000	50	4.00e-04	0.015	0.200	0.200	0.000	0.000	1	no
07_both_strong	Both strong	1000	50	4.00e-04	0.015	0.500	0.500	0.000	0.000	1	no
08_realistic	Realistic	2500	100	4.00e-04	0.015	0.150	0.350	0.000	0.000	1	no
09_garch_mild	GARCH mild	1000	50	4.00e-04	0.015	0.200	0.200	0.050	0.900	1	no
10_garch_strong	GARCH strong	1000	50	4.00e-04	0.015	0.300	0.400	0.100	0.850	1	no
11_multifactor	Multi-factor K=3	1000	50	4.00e-04	0.015	0.200	0.300	0.000	0.000	3	no
12_regime	Regime-switching	2000	50	4.00e-04	0.015	0.150	0.250	0.000	0.000	1	yes
13_high_ar1	High-persistence AR(1)	1000	50	4.00e-04	0.015	0.700	0.300	0.000	0.000	1	no
14_garch_highvol	High-volatility GARCH	1000	50	4.00e-04	0.015	0.200	0.300	0.150	0.800	1	no

Table 20. Coverage of the date-level Sharpe target by alternative interval procedures.

DGP	blocked	hac_delta	iid	row_naive	stationary
01_iid	0.947	0.952	0.951	0.000	0.947
02_ser_mild	0.938	0.943	0.898	0.000	0.936
03_ser_strong	0.921	0.934	0.734	0.000	0.919
04_xs_mild	0.950	0.950	0.952	0.073	0.950
05_xs_strong	0.934	0.938	0.942	0.262	0.938
06_both_mod	0.937	0.939	0.898	0.088	0.936
07_both_strong	0.934	0.943	0.771	0.178	0.935
08_realistic	0.937	0.941	0.901	0.107	0.937
09_garch_mild	0.926	0.940	0.877	0.110	0.930
10_garch_strong	0.926	0.948	0.862	0.199	0.927
11_multifactor	0.937	0.947	0.895	0.225	0.940
12_regime	0.929	0.934	0.890	0.306	0.929
13_high_ar1	0.899	0.933	0.562	0.137	0.907
14_garch_highvol	0.940	0.948	0.895	0.193	0.941

Note: Row-naive intervals are evaluated against the economically relevant date-level Sharpe target, not against their own row-distribution Sharpe target. Low row-naive coverage therefore records target mismatch and false precision at the portfolio boundary.

Table 21. Row-naive interval behavior when evaluated against the date-level Sharpe target.

DGP	Name	row coverage	bias vs date SR	CI width	mean SE	n
01_iid	IID baseline	0.000	-2.567	0.278	0.071	1000
02_ser_mild	Serial mild	0.000	-2.567	0.278	0.071	1000
03_ser_strong	Serial strong	0.000	-2.568	0.278	0.071	1000
04_xs_mild	Cross mild	0.073	-0.474	0.278	0.071	1000
05_xs_strong	Cross strong	0.262	-0.153	0.278	0.071	1000
06_both_mod	Both moderate	0.088	-0.503	0.278	0.071	1000
07_both_strong	Both strong	0.178	-0.145	0.278	0.071	1000
08_realistic	Realistic	0.107	-0.277	0.124	0.032	1000
09_garch_mild	GARCH mild	0.110	-0.473	0.278	0.071	1000
10_garch_strong	GARCH strong	0.199	-0.268	0.278	0.071	1000
11_multifactor	Multi-factor K=3	0.225	-0.327	0.278	0.071	1000
12_regime	Regime-switching	0.306	-0.092	0.197	0.050	1000
13_high_ar1	High-persistence AR(1)	0.137	-0.338	0.278	0.071	1000
14_garch_highvol	High-volatility GARCH	0.193	-0.325	0.278	0.071	1000

Note: These are not ordinary row-estimator coverage rates; they evaluate the row-naive interval against the economically relevant date-level target.

Table 22. Sampling-boundary stress tests under a null design with no true edge.

Panel	Setting	Row rej.	HAC rej.	Row mean SE	HAC mean SE
A. Same-date correlation (N=50)	rho=0.000	0.097	0.056	0.071	0.598
A. Same-date correlation (N=50)	rho=0.100	0.310	0.064	0.071	0.598
A. Same-date correlation (N=50)	rho=0.300	0.354	0.044	0.071	0.599
A. Same-date correlation (N=50)	rho=0.350	0.368	0.071	0.071	0.597
A. Same-date correlation (N=50)	rho=0.500	0.409	0.047	0.071	0.597
A. Same-date correlation (N=50)	rho=0.700	0.408	0.060	0.071	0.599
B. Selected count (rho=0.35)	N=5	0.192	0.058	0.225	0.598
B. Selected count (rho=0.35)	N=25	0.308	0.052	0.100	0.600
B. Selected count (rho=0.35)	N=50	0.368	0.071	0.071	0.597
B. Selected count (rho=0.35)	N=100	0.421	0.056	0.050	0.598

Note: Row-naive overrejection rises with same-date dependence and selected-count pressure, whereas date-level HAC remains close to nominal size.

Table 23. Monte Carlo rejection rates under null designs with no true edge.

Null design	Row-naive	Date-IID bootstrap	HAC-delta	Moving-block bootstrap	Stationary bootstrap	Romano-Wolf
Dependent null	0.352	0.094	0.066	0.070	0.067	0.068
GARCH null	0.351	0.098	0.063	0.070	0.068	0.067
IID null	0.062	0.060	0.060	0.061	0.062	0.060

Note: Each entry is the fraction of 1000 Monte Carlo replications rejected at nominal 5 percent size.

The main null designs use T=1000 dates and N=50 entities. Dependent null combines same-date and serial dependence; GARCH null adds conditional heteroskedasticity; IID null removes both. Row-naive treats selected asset-date rows as independent; the other procedures operate on the date-level portfolio return series.

Table 24. Rejection rates across true annualized Sharpe values.

index	true_sharpe	date_iid	hac_delta	moving_block	romano_wolf	row_naive	stationary
0.000	0.000	0.081	0.053	0.058	0.055	0.371	0.056
1.000	0.094	0.105	0.071	0.072	0.072	0.424	0.077
2.000	0.189	0.139	0.088	0.094	0.095	0.495	0.092
3.000	0.378	0.219	0.151	0.160	0.162	0.610	0.162
4.000	0.755	0.459	0.354	0.373	0.376	0.797	0.380
5.000	1.511	0.858	0.788	0.805	0.800	0.985	0.798
6.000	2.833	1.000	1.000	1.000	1.000	1.000	1.000